

Simon Lin

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SUMMARY

Software Developer with experience building full-stack applications, automation tools, and data systems across the frontend, backend, and cloud. Proven track record of shipping features that improve performance and engineering efficiency. Passionate about system design, scalable code and solving problems that have real user impact.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, C#, TypeScript, Java, SQL

Libraries & Frameworks: React, Vue.js, .NET, Three.js, Flask, OpenCV, Selenium, Postman

Developer Tools: AWS, Azure Cloud, Git, Jira, Docker, Azure DevOps, GitHub

EXPERIENCE

Software Engineer Co-op

May 2025 – Dec 2025

Magna International

Newmarket, ON

- Developed and deployed full-stack web application (**Python + Flask + React**) using **AWS** to accelerate Finite Element Analysis (FEA) processes to support a team of global engineers.
- Implemented a 3D FEA model viewer using **Three.js**, to visualize complex tensile stresses in a web browser.
- Built a CATIA automation tool, reducing manual data entry time by 70% in CAD files and improving efficiency for cross-functional engineering teams.

Software Engineer Co-op

Apr 2024 – Aug 2024

SOTI

Mississauga, ON

- Developed new web application features in **C#** and **.NET**, adhering to technical and customer specifications.
- Implemented **RESTful** APIs to improve response times between client applications and backend services by **35%**.
- Designed and implemented relational database schemas using **Microsoft SQL Server**, normalizing data models to reduce redundancy and improved query performance across 5+ core product tables.
- Collaborated in an Agile CI/CD pipeline, participating in weekly planning and code reviews to ship features.

Software Automation Engineer Co-op

Sep 2023 – Apr 2024

SOTI

Mississauga, ON

- Created automated test plans, test cases and unit tests in **C#**, **Selenium** and **Microsoft Azure DevOps**.
- Discovered and resolved over **155** defects and performance issues across product UI and dashboards.
- Lowered SQL database retrieval time by **27%** by optimizing legacy queries to clean, transform & aggregate data.

Python Coding Instructor

Jul 2021 – Aug 2021

The STEAM Project

Richmond Hill, ON

- Taught 20 summer camp students from grades 7-8 a creative curriculum centered on Python fundamentals.
- Collaborated with co-instructors to create an educational and interactive coding curriculum on a weekly basis.

EXTRACURRICULARS

Embedded Firmware Engineer

Sep 2023 – Dec 2023

TMU Formula SAE

Toronto, ON

- Developed firmware in C for Arduino microcontrollers to control and poll vehicle subsystems.
- Developed a telemetry dashboard with Python libraries to support testing workflows and enable real time diagnostics.

EDUCATION

Bachelor of Science - Computer Science with Co-op

Sep 2021 – Present

Toronto Metropolitan University

Toronto, ON

- Relevant Courses: Software Engineering, Data Structures, Artificial Intelligence, Web Development, OS.

PROJECTS

NBAction | [GitHub Repository](#)

- Developed and trained a deep-learning model capable of detecting basketball actions to analyze live basketball games.
- Optimized a video processing pipeline using dynamic frame sampling algorithms to reduce live video latency by 68%.
- Performed model fine-tuning to achieve an overall detection rate of 92% across all object classes.